



Distribution of Offense Types Across NYC Boroughs

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What We Learned: Key Findings at a Glance

Research Objective

Assess whether offense patterns vary across NYC boroughs and what those differences imply for location-aware health insights.

Why This Matters Now

Uneven exposure patterns suggest **localized differences in injury and emergency care demand**, reinforcing the value of spatially informed health analytics.

Geographic Variation

Offense frequency varies meaningfully across NYC boroughs, revealing spatial patterns in exposure risk.

Shared Categories, Different Intensities

Boroughs share similar offense types, but **exposure intensity differs by location**.

Early Directional Signals

Even with limited data, **directional spatial signals emerge** that are relevant for health risk modeling.

Why Spatial Variation Matters for Health Tech

Core Insight

Geographic variation in public safety data can serve as an **early indicator of localized health risk**. Emergency department demand is not evenly distributed across a city.

Operational Relevance

Risk stratification improves when models incorporate **where incidents occur**, not just what occurs. This enables more precise resource allocation and intervention planning.

Health Tech Applications



Demand Forecasting

Predict emergency service needs by location and time



Risk Stratification

Location-aware population health modeling



Preventive Interventions

Surveillance-oriented public health strategies

Boroughs Differ in Offense Exposure

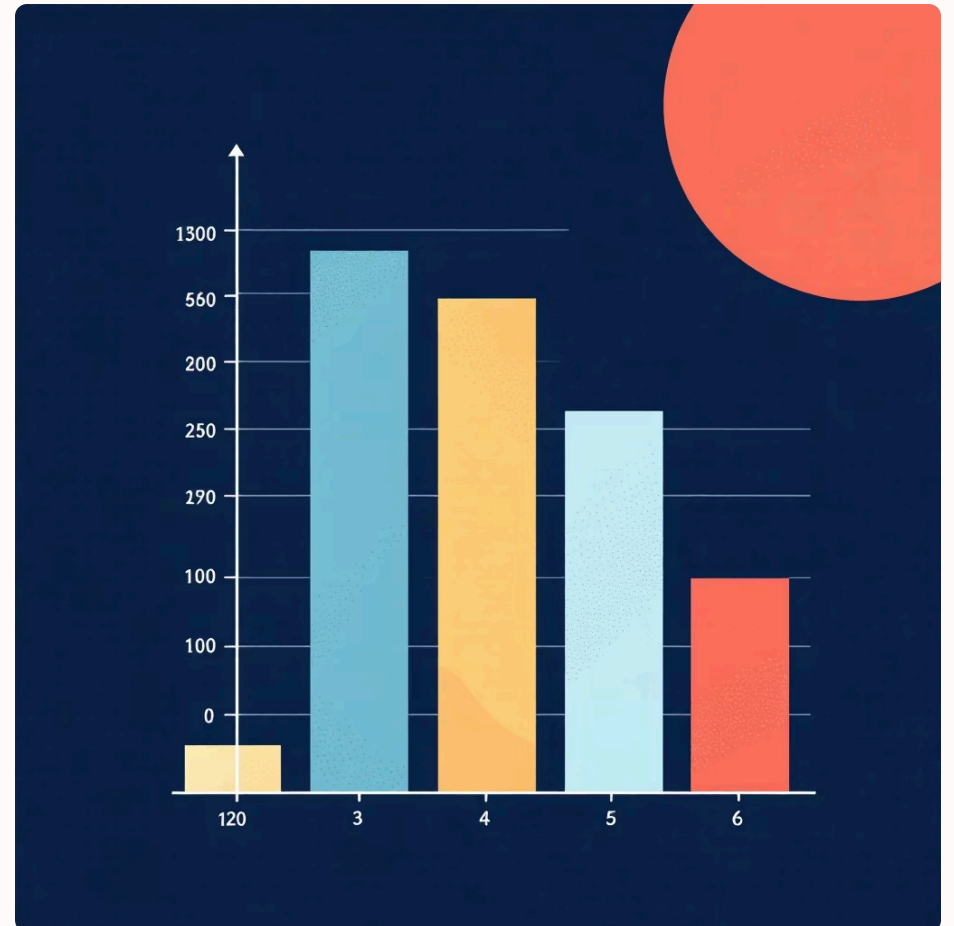
Primary Finding

Boroughs show clear differences in the **volume of reported offenses**, indicating uneven exposure risk across NYC's geography.

Evidence

- Brooklyn exhibits the highest offense count in the dataset
- Staten Island consistently shows the lowest volume
- Other boroughs fall between these extremes

📄 **Health Implication:** These differences suggest that health system demand and injury risk may vary by location, even within the same city.



Similar Offense Types, Different Intensities



Consistent Categories

The same offense categories appear across all boroughs, indicating compositional stability.



Variable Frequency

The **frequency of those offenses differs substantially by location**, revealing exposure gradients.

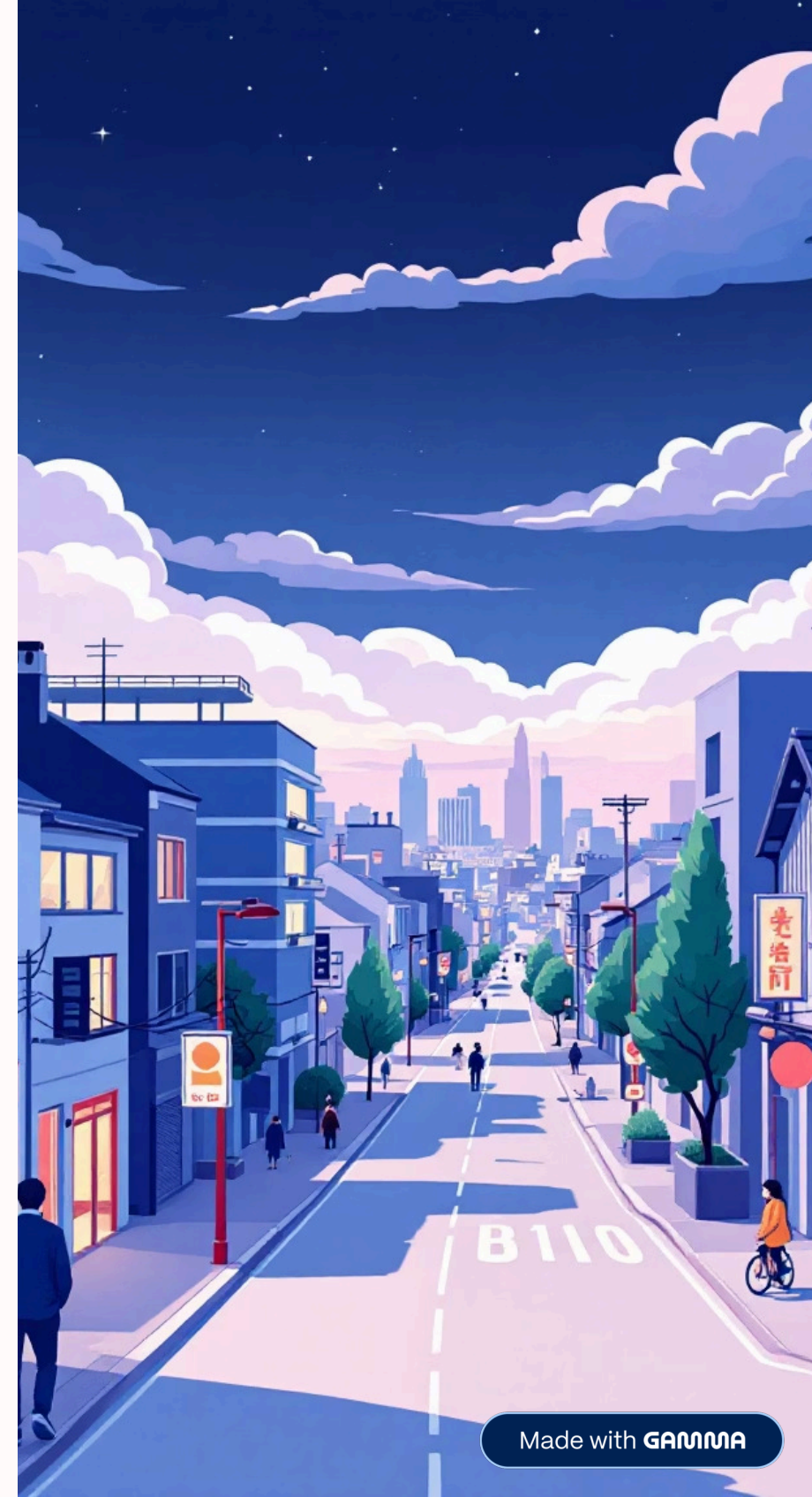


Modeling Implications

Predictive models may generalize across regions, but resource allocation and intervention strategies should remain **location-specific**.

What This Means for Implementation

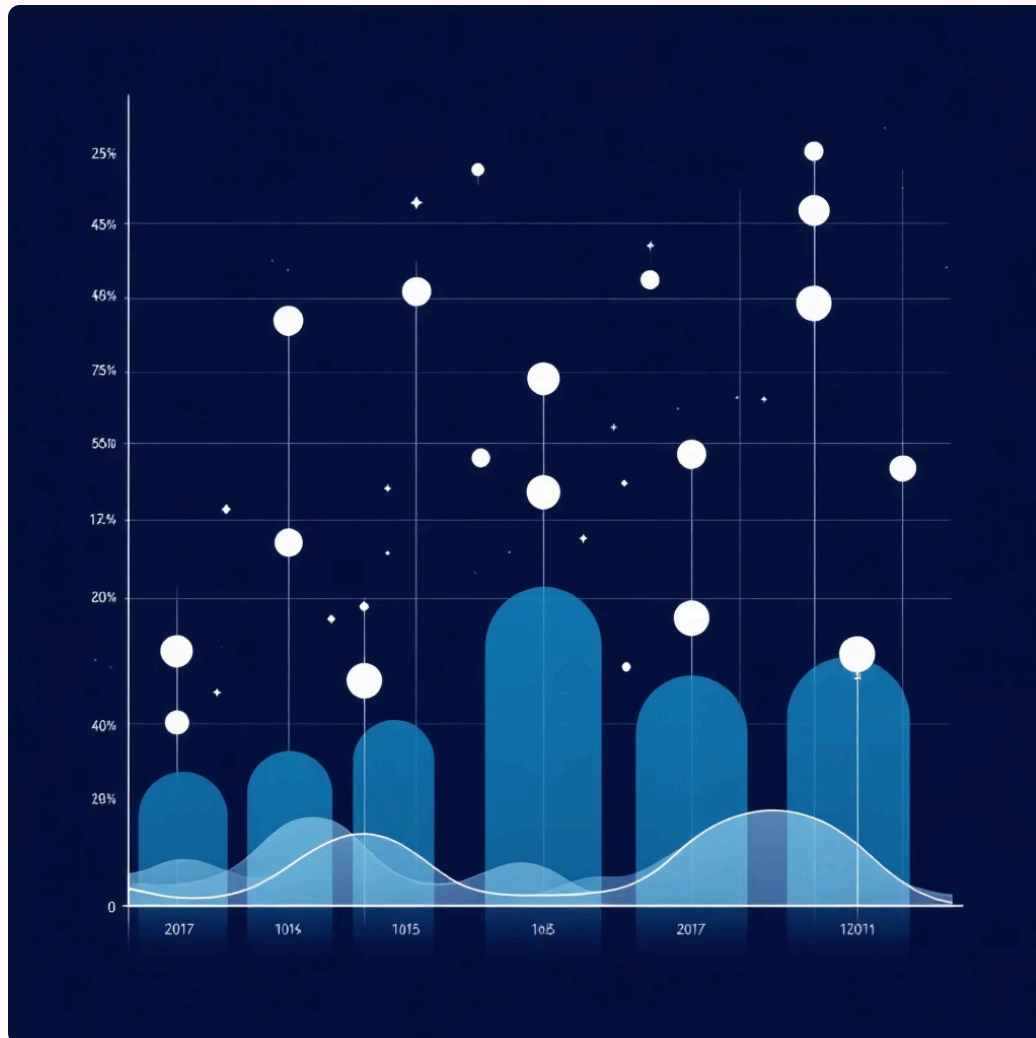
Offense composition is stable across boroughs, but exposure intensity is not. This pattern supports building scalable models while maintaining localized deployment strategies.



What the Statistics Do (and Don't) Tell Us

Statistical Outcome

No statistically significant association detected between offense type and borough. This result likely reflects **small sample size and sparse categories**, not absence of structure.



Correct Interpretation

Lack of statistical significance $\neq$ lack of signal

Findings should be interpreted as **exploratory and directional**. Spatial patterns appear consistently across multiple views of the data.

Why This Still Matters

Statistical power—and confidence—would increase with broader datasets. Current results establish proof-of-concept for spatial structure.

Strategic Implications & Next Steps

What These Findings Enable at Scale

Location-Aware Risk Modeling

Build spatially-informed injury risk prediction systems

Demand Forecasting

Improve emergency department resource planning

Surveillance Systems

Enable data-driven public health monitoring

Data Expansion Opportunities

01

Incorporate temporal variables to detect trends

02

Normalize offense counts by population density

03

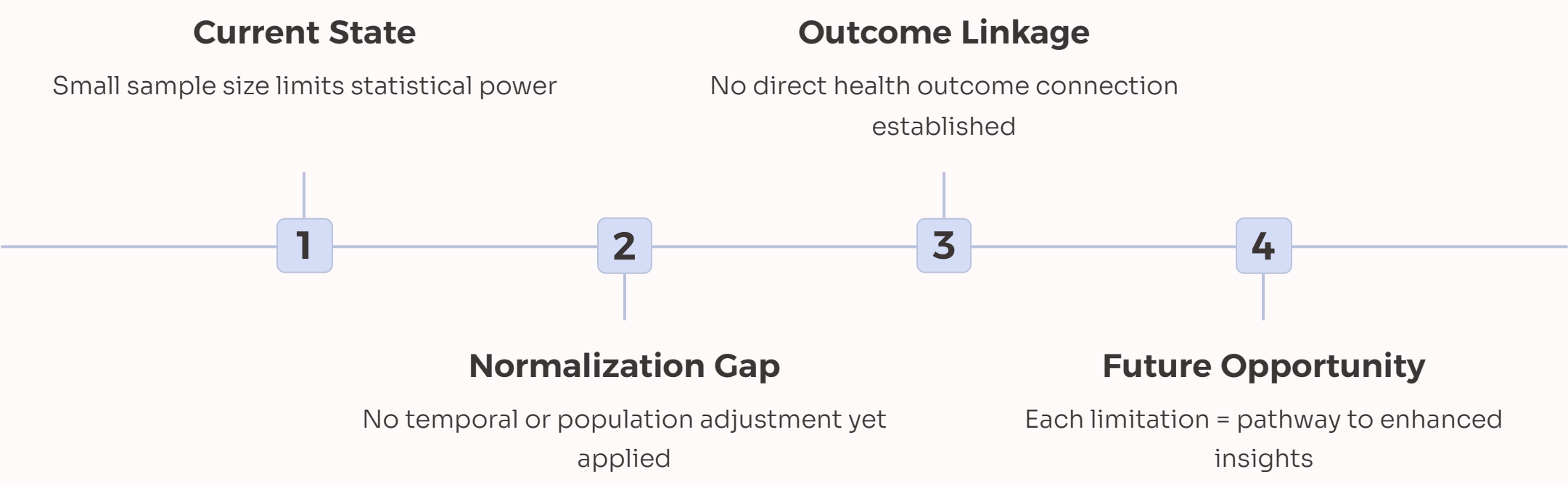
Link offense data with health outcome datasets (e.g., ER visits)

 **Executive Takeaway:** This analysis demonstrates **proof-of-concept spatial signal** with clear potential to inform scalable health tech solutions.



Limitations as a Roadmap

Each constraint represents a clear path to higher-impact insights. Expanded data would unlock stronger inference and product value.



Why These Are Opportunities

- Temporal expansion enables trend detection
- Population normalization improves comparability
- Health linkage validates predictive value
- Broader datasets strengthen causal inference

